A SECTION VANISHING ON AN OPEN CONTAINING THE ASSOCIATED POINTS VANISHES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Associated points behave for short exact sequences. Lemma lemma-ses-ass, used at proof lines 210-212. Gives Ass(F') contained in Ass(F).
- Associated points lie in the support. Lemma lemma-ass-support, used at proof lines 212-213. Gives Ass(F') contained in Supp(F').
- A quasi-coherent module with no associated points is zero. Lemma lemma-ass-zero, used at proof lines 214-215. Concludes F' = 0.

Source reference. Original lemma 05AD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6846–6846. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05AG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6849–6849. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05AE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6847–6847. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a locally Noetherian scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. If $U \subset X$ is open and $Ass(\mathcal{F}) \subset U$, then $\Gamma(X, \mathcal{F}) \to \Gamma(U, \mathcal{F})$ is injective.*

Proof. Let $s \in \Gamma(X, \mathcal{F})$ be a section which restricts to zero on U. Let $\mathcal{F}' \subset \mathcal{F}$ be the image of the map $\mathcal{O}_X \to \mathcal{F}$ defined by s. Then $\text{Supp}(\mathcal{F}') \cap U = \emptyset$. On the other hand, $\text{Ass}(\mathcal{F}') \subset \text{Ass}(\mathcal{F})$ by Lemma 05AE. Since also $\text{Ass}(\mathcal{F}') \subset \text{Supp}(\mathcal{F}')$ (Lemma 05AD) we conclude $\text{Ass}(\mathcal{F}') = \emptyset$. Hence $\mathcal{F}' = 0$ by Lemma 05AG. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.